

RAKESH SHARMA

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CLOUD DATA ARCHITECT (ANALYTICS & AI)

Principal Data Architect with 17+ years leading enterprise data architecture and cloud (AWS) implementation for financial services organizations. Proven track record designing and delivering large-scale Lakehouse and data platform strategies on AWS (S3, Glue, EMR, Redshift, Lake Formation, DMS, Kinesis), leading cross-functional engineering teams, and driving multi-year cloud modernization roadmaps for banking, insurance, and asset management clients. Deep expertise in data security, RBAC/governance, regulatory compliance (OSFI, PIPEDA, SOX), and executive-level architecture leadership.

CORE COMPETENCIES

Architecture: Lakehouse & Data Mesh Design | Data Vault 2.0 | Dimensional Modeling | Multi-Zone Data Lake Design | Infrastructure-as-Code (Terraform, AWS CDK) | CI/CD

Data Integration: dbt | Fivetran (HVR) | AWS Glue | Azure Data Factory | Informatica | Airflow | Kafka | Kinesis | Spark Streaming | Batch & Real-Time Pipelines

Cloud Platforms: AWS (S3, Glue, EMR, Lambda, Kinesis, SageMaker, Bedrock, CDK) | Azure (ADF, Databricks, Synapse, ADLS, Azure AD) | GCP (BigQuery, Dataflow)

Governance & Security: Data Masking | Encryption | Lineage Tracking | Metadata Management | Great Expectations | Data Mesh | PIPEDA | SOX | OSFI Compliance

BI & Analytics: Power BI | Tableau | Dimensional Modeling | Self-Service Reporting | Prometheus & Grafana Monitoring

AI/ML Enablement: LangChain | LlamaIndex | CrewAI | Snowflake Cortex | OpenAI | Cohere | SageMaker | LLMops/MLOps | RAG | Fine-Tuning (QLoRA, LoRA)

Languages: SQL (Advanced Query Optimization) | Python | Scala | Shell

PROFESSIONAL EXPERIENCE

Cloud Data Architect — Banking Labs – Bank of Montreal

May 2025 – Present

- Defined and drove the enterprise data architecture strategy and multi-year roadmap for cloud modernization on AWS, aligning platform investments with business objectives across capital markets, retail banking, and wealth management.
- Architected a unified enterprise Lakehouse platform on AWS (S3, Glue, EMR, Redshift Serverless, Lake Formation) consolidating 40+ siloed data warehouses into a single governed platform, reducing total cost of ownership by an estimated 35%.
- Led end-to-end design of the AWS data lake: multi-zone S3 architecture (Landing, Staging, Standard, Archive), KMS encryption at rest, PGP encryption for high-risk data, and 7-year regulatory retention aligned to requirements.
- Architected batch and real-time ingestion pipelines using AWS DMS (700+ tables), Kinesis/Kinesis Firehose, EventBridge, Step Functions, and Lambda for fraud detection, trade surveillance, and transaction monitoring at 2M+ events/sec with sub-second latency.
- Designed and implemented enterprise RBAC and data security architecture using AWS Lake Formation, IAM fine-grained access policies, and Secrets Manager, enforcing least-privilege access and column/row-level data masking for PII across 12 business domains.
- Established infrastructure-as-code standards using Terraform and AWS CDK across all data platform components, achieving fully automated provisioning, environment parity (Dev/Test/Prod), and CI/CD integration with GitHub Actions and Jenkins.
- Led a cross-functional team of 15+ data engineers, ML engineers, and cloud architects — setting technical standards, hiring rubrics, and code-review practices, and establishing a Data & AI Center of Excellence.
- Delivered executive-level architecture reviews and roadmap presentations to CTO/CIO leadership, translating complex AWS platform initiatives into business-impact narratives to secure incremental investment funding.

- Designed a scalable data quality and observability framework (Great Expectations, dbt tests, CloudWatch) enforcing automated validation and anomaly detection across critical financial data pipelines on AWS.

Senior Data & AI Architect — Fidelity International Canada

January 2020 – April 2025

- Led enterprise-scale data migration from SQL Server, DB2, and Oracle Data Warehouse to Snowflake, implementing Lakehouse architecture with star schema dimensional models that enabled efficient cross-functional analytics with optimized query performance.
- Designed and optimized Snowflake warehouse configurations including clustering strategies, materialized views, and compute/storage separation to maximize cost efficiency and query performance for enterprise reporting workloads.
- Architected comprehensive data pipeline solutions implementing dbt for transformation and Fivetran HVR for change data capture, enabling multi-modal data ingestion (batch and continuous streaming) from DB2 and SQL Server sources into Snowflake.
- Built automated ingestion pipelines using AWS Glue, Lambda, Kinesis Firehose, and DynamoDB alongside Azure Data Factory, Azure Databricks, and Synapse to feed Snowflake with data from ERP, CRM, and operational systems.
- Developed end-to-end CI/CD pipelines on Terraform for the Snowflake-based analytics platform across AWS and Azure, ensuring automated provisioning and repeatable deployments.
- Developed an advanced monitoring solution tracking key performance metrics of Snowflake, HVR, Power BI, and dbt Cloud using Prometheus and Grafana, enabling proactive cost management and warehouse consumption optimization.
- Designed and implemented data archival strategies within Snowflake including automated ETL processes for data lifecycle management, improving storage efficiency and query performance on active datasets.
- Architected and implemented comprehensive data lake security framework across Snowflake, AWS, and Azure, featuring RBAC with granular permissions integrated with Azure AD, network perimeter security, and dynamic data masking for sensitive information.
- Implemented GenAI Agentic RAG to automate purging of expired data across teams, streamlining data governance and compliance using both closed and open-source LLM technology integrated with Snowflake Cortex.
- Established seamless integration between Snowflake and enterprise BI tools, with focus on Power BI for organization-wide analytics, enabling self-service reporting for Finance, Operations, and Business stakeholders.
- Developed GenAI solutions including a Data Catalog (Data Crawler) and Document Parser on Snowflake Cortex, providing Data Governance and Business teams with automated insights into enterprise data assets.

EARLIER CAREER

Technical Consultant — TCS – Economical Insurance & Enbridge Gas

January 2017 – January 2020

- Architected cloud data lakes on AWS, Azure, and GCP for insurance and energy clients; led data ingestion frameworks using Airflow, Data Fusion, and Talend; integrated SAP HANA with cloud data lakes for real-time and batch pipelines.

Senior Software Engineer — Pitney Bowes

January 2016 – January 2017

- Led migration of the enterprise data warehouse to AWS EMR; built Spark Streaming workflows with Kafka and Cassandra; developed dashboards and ML-based customer segmentation.

Senior Analyst Programmer — Fidelity International

December 2012 – January 2016

Systems Engineer — Infosys Limited

August 2009 – December 2012

EDUCATION

Bachelor of Technology, Computer Science

Guru Nanak Dev University

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate – AWS
- AWS Certified Data Analytics – Specialty – AWS
- Snowflake SnowPro Core – Snowflake
- Google Cloud Professional Data Engineer – Google Cloud
- Cloudera Apache Hadoop Certified – Cloudera
- MCSD – Certified Spark Developer